

Polymethyl Methacrylate (PMMA): Structure, Processing, and Applications in Biomedicine and Semiconductor Engineering

PMMA, which stands for poly (methyl methacrylate), is a class of thermoplastic polymers characterized by transparency and stability; however, the uses of PMMA in orthopedics and semiconductor industry have been explored independently. This review article highlights the role of PMMA in orthopedic surgery and semiconductor lithography as applications that benefit from the same physical characteristics and structures of PMMA but utilize their properties to different degrees. Specifically, PMMA bone cement is focused on mechanical stability and biocompatibility whereas the focus of PMMA resist is on high-resolution imaging and chain breakage under electron-beam irradiation. Notwithstanding drawbacks associated with thermal necrosis in orthopedic applications and poor etch ability in semiconductor lithography, PMMA is still considered a versatile platform material due to continuous modifications with respect to the addition of bioactive molecules and hybridization with other photoresist materials.

Chemical Structure and Core Properties

PMMA's value lies in its versatile property-structure relationships. Its chemical inertness and high viscosity allow use as bone cement in biomedical applications and as a resist in semiconductor fabrication. PMMA is a linear addition polymer, synthesized by free-radical polymerization of methyl methacrylate monomers. Its backbone is a saturated carbon chain, with pendant ester side groups for optical clarity and chemical resistance. The empirical formula for the repeat unit is $C_5H_8O_2$. Molecular mass, determined by the degree of polymerization, strongly influences mechanical and rheological properties (EBSCO Research Starters, 2024). The main defining property of PMMA is its optical clarity. PMMA transmits around 92% of visible light, outperforming glass and most transparent polymers. This clarity is due to PMMA's amorphous, non-crystalline structure, which does not scatter light. Its refractive index of about 1.49 makes it suitable for optical applications such as lenses and light guides. Mechanically, PMMA is a rigid, glassy solid at room temperature with a tensile strength of 50-75 MPa and Young's modulus of 2.4-3.3 GPa. PMMA's glass transition temperature (T_g) spans from 85°C to 165°C, influenced by molecular weight and formula. Above T_g , PMMA softens, making this thermal response significant for its use in both biomedical and semiconductor applications.

Chemically, PMMA is resistant to dilute acids, alkalis, and many aqueous solutions. It is vulnerable to ketones, esters, and aromatic solvents. This selective solubility is not a weakness; it is a feature used in lithographic processing. PMMA films must dissolve cleanly in specific developer solutions after electron beam or UV exposure. In biomedical settings, this same chemical stability ensures that implanted PMMA resists degradation in the physiological environment over long periods. PMMA is inherently biocompatible in its cured form. It does not create significant immune responses when implanted in the body. In the 1940s, ophthalmologists noticed PMMA fragments from shattered aircraft canopies caused a slight tissue reaction in

injured pilots. This discovery led to PMMA's later use in intraocular lenses, bone cements, and dental prosthetics (Frazer et al., 2005, pp. 629-639).

PMMA's properties are highly tunable and depend significantly on synthesis conditions. Variations in molecular weight, polymerization mechanism, additive type, and mixing procedures substantially influence its final characteristics, which are critical for meeting diverse application requirements. Molecular weight is the primary factor affecting PMMA's mechanical response. Higher-molecular-weight chains increase tensile strength, toughness, and glass transition temperatures, but also raise melt viscosity, complicating flow-dependent applications such as surgical bone cement injection or spin-coating thin films. Engineers optimize molecular weight for each application, balancing mechanical performance and manufacturability. Bone cement formulations typically use lower-molecular-weight PMMA powders, which are mixed with liquid MMA monomer to form an injectable matrix. In contrast, PMMA resists for lithography are specified at higher molecular weights to ensure imaging resolution and detection. Several MMA polymerization routes exist, each tailored to different end-use properties. Bulk polymerization yields optically pure PMMA but is challenging to control due to strong exothermicity. Solution polymerization in organic solvents provides more uniform reaction kinetics and is often used for PMMA in microfabrication. Suspension and emulsion polymerization produce PMMA bead particles for orthopedic bone cements. The selected synthesis pathway directly affects the purity and intended function of the PMMA product. Biomedical and semiconductor applications rarely employ unmodified PMMA. In orthopedics, specific additives are incorporated to tailor radiopacity and clinical effectiveness. Radiopacifiers such as barium sulfate or zirconium dioxide enhance X-ray visibility, antibiotics like gentamicin reduce infection risk, plasticizers decrease brittleness, and crosslinking agents increase compressive strength.

Application requirements in semiconductors and medicine clarify why PMMA is formulated differently. In semiconductors, PMMA resists are dissolved in solvents at precise concentrations for controlled thin film deposition and reliable lithography. High purity is vital to prevent defects. In bone cement, mixing powder and monomer triggers an exothermic reaction: the generated heat can reach levels (40°C to 120°C) that threaten bone health above 47°C. This thermal surge, unique to medical settings, risks tissue damage and drives the development of improved formulas with less monomer and lower reaction heat. By contrast, semiconductor processing uses PMMA at room or low temperatures, presenting fundamentally different thermal challenges despite the shared base material. PMMA has established itself as one of the most widely used synthetic polymers in medicine, with applications encompassing orthopedic surgery, dental restoration, and ophthalmology. Its biocompatibility, chemical stability, and modifiable mechanical properties make it uniquely suited for implantation in the human body, where it must perform reliably for years or decades in a chemically complex, mechanically demanding environment. The most clinically significant biomedical application of PMMA is its use as a bone cement in orthopedic procedures. First introduced into orthopedic surgery by Sir John

Charnley in the 1950s for total hip arthroplasty, PMMA bone cement serves as a grout rather than a true adhesive; it fills the irregular space between a prosthetic implant and the surrounding bone, distributing mechanical loads and securing the implant in place. The cement is prepared intraoperatively by mixing a powder component containing pre-polymerized PMMA beads, a radiopacifier, and an initiator with a liquid monomer component of methyl methacrylate. The resulting paste is injected into the bone cavity and allowed to polymerize in situ, conforming to the geometry of the surgical site before hardening into a rigid fixation material.

PMMA bone cement is used in a range of orthopedic procedures, including total hip and knee arthroplasty, vertebroplasty, and kyphoplasty, minimally invasive procedures in which cement is injected directly into fractured vertebral bodies to restore physical stability and relieve pain. In each of these applications, the cement must meet strict mechanical requirements: compressive strength sufficient to withstand physiological loads, adequate stiffness to match the modulus of the surrounding bone, and sufficient working time to allow the surgeon to position the implant before hardening. In its fully cured form, PMMA is generally well tolerated by the body. It does not provoke significant acute immune responses and has demonstrated durable stability in implanted applications that last for decades. However, the biological response to PMMA is not entirely benign. Residual unreacted MMA monomer, which is present in freshly mixed bone cement before polymerization is complete, is cytotoxic and can cause local tissue irritation. Systemic absorption of MMA monomer during cement implantation has been associated with transient cardiovascular effects in some patients, a phenomenon sometimes referred to as bone cement implantation syndrome. Additionally, wear debris generated at the cement-implant interface over time can trigger a chronic inflammatory response, causing aseptic loosening, one of the leading causes of long-term implant failure. Though it has a long clinical track record, PMMA bone cement has several well-documented limitations. The exothermic polymerization is a primary concern (Placek, L., Perry, D., Towler, M., & Wren, A., 2025). Table 2 compares the temperature range of PMMA polymerization with the threshold for bone thermal necrosis.

Condition	Temperature (°C)
PMMA bone cement peak (range)	40 – 120
Bone thermal necrosis threshold	>47

Table 2. Peak polymerization temperature of PMMA bone cement vs. thermal necrosis threshold of bone. Data from Placek et al. (2025).

Mechanically, PMMA bone cement is brittle compared with cortical bone and cannot bond chemically to bone surfaces, relying entirely on mechanical interlock for fixation. Over time, cyclic loading can generate microcracks within the cement mantle, which propagate and eventually contribute to implant loosening. PMMA does not promote bone ingrowth or remodeling, so the cement-bone interface does not improve over time, as a cementless implant interface might through osseointegration. These limitations have driven substantial research into

modified PMMA formulations. Incorporation of bioactive fillers, such as hydroxyapatite, bioactive glass, and carbon nanotubes, has been explored to improve biological response at the cement-bone interface. Antibiotic-loaded cements, which release therapeutic concentrations of antibiotics locally over time, have become standard practice in revision surgery and high-infection-risk cases. These developments reveal an ongoing effort to extend and enhance the clinical utility of a material that, despite its limitations, remains irreplaceable in many surgical contexts. As summarized in Table 1, the two fields emphasize different properties.

Feature	Biomedical (Bone Cement)	Semiconductor (E-beam Resist)
Primary Function	Fixate Implants, fill bone voids	Transfer nanoscale patterns
Molecular Weight	Lower (easier injection)	Higher (sharper features)
Key Advantage	Biocompatibility, stability	High resolution, chain scission
Main Limitation	Thermal necrosis, aseptic loosening	Poor etch resistance, low sensitivity
Typical Additives	Barium sulfate, antibiotics	Metal oxides (EUV use)
Timescale	Years to Decades	Hours

Specifications of PMMA in the Semiconductor Industry

While PMMA's role in medicine is well established, its contribution to semiconductor manufacturing represents an equally significant but less widely recognized application domain. In the semiconductor industry, PMMA primarily serves as a photoresist, a light- or electron-sensitive polymer film used to transfer circuit patterns onto silicon wafers during the fabrication of integrated circuits. Its high-precision capability, straightforward processing, and well-characterized sensitivity to electron-beam exposure have made it the standard resist material for research-scale nanofabrication and a critical reference point against which newer resist materials are benchmarked. To understand PMMA's role in semiconductor manufacturing, it is necessary to first understand the basic principles of lithographic patterning. Lithography is the process by which a geometric pattern, representing transistors, interconnects, or other circuit features, is transferred from a design file onto the surface of a silicon wafer. The process starts with the deposition of a thin, uniform resist film on the wafer surface, frequently achieved by spin-coating a solution of PMMA in an organic solvent, such as anisole. The resist-coated wafer is then exposed to a patterning energy source, ultraviolet light, an electron beam, or extreme ultraviolet radiation, which chemically modifies the exposed regions of the resist. In the case of PMMA, exposure causes chain scission, the breaking of polymer backbone bonds, which reduces the molecular weight of the exposed material and increases its solubility in a developer solution. The

e-beam lithography process using PMMA is illustrated in Figure 1, which shows a typical workflow from exposure to final pattern transfer.

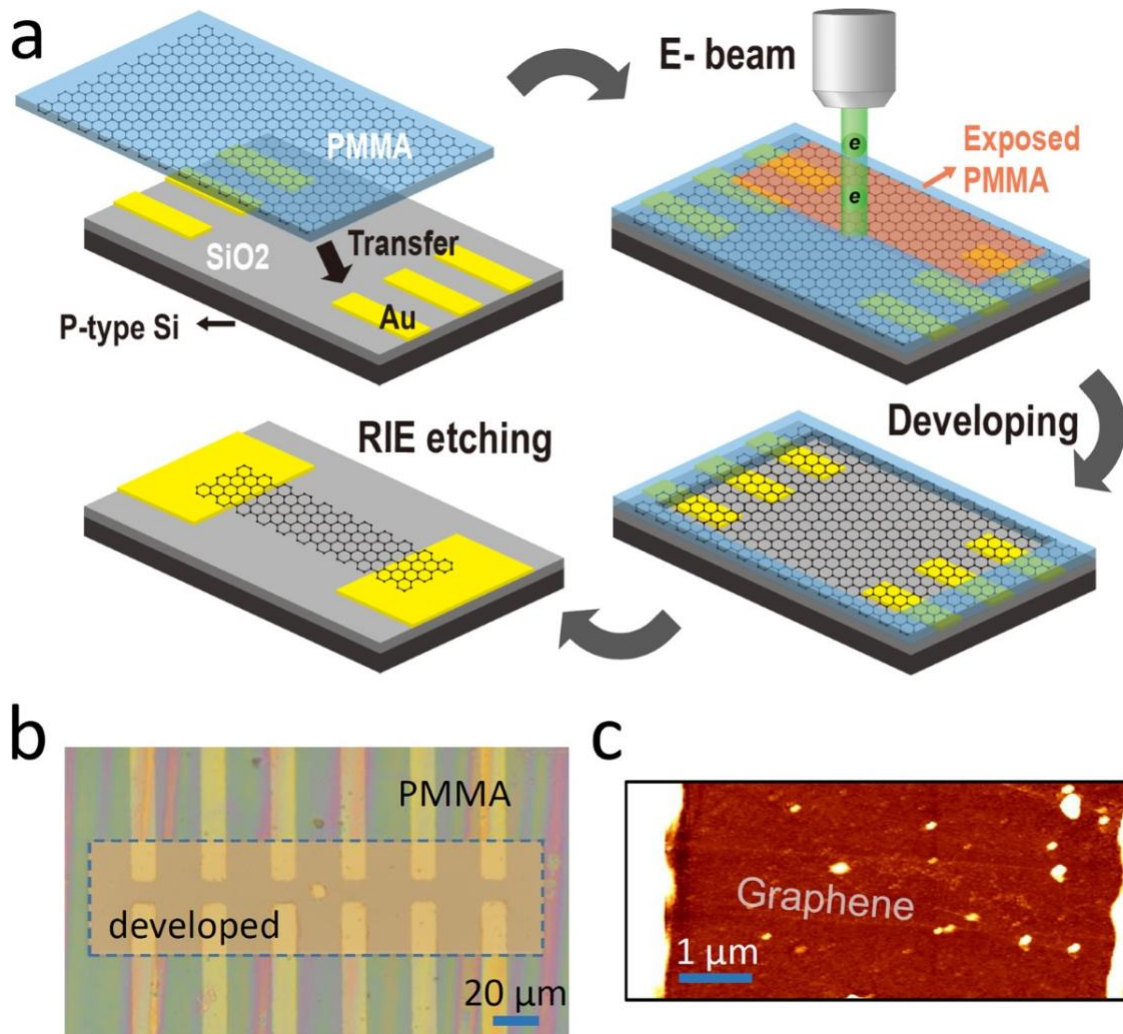


Figure 1. PMMA as an e-beam resist for pattern transfer. Adapted from Son et al. (2017).

When the wafer is subsequently immersed in developer, the exposed regions dissolve, leaving a patterned resist mask. This mask protects the underlying material during subsequent etching or deposition steps, ultimately defining the physical features of the circuit. PMMA's most important role in semiconductor processing is as a positive-tone electron beam (e-beam) resist. In e-beam lithography, a focused electron beam is scanned across a resist-coated wafer to write patterns directly, without a physical mask. This direct-write capability makes e-beam lithography extremely valuable for research applications, photomask fabrication, and the definition of nanoscale features beyond the resolution limits of conventional optical lithography.

PMMA is exceptionally well-suited to e-beam lithography for several reasons. Its sensitivity to electron beam exposure is high and well characterized, allowing accurate control of dose and pattern fidelity. The chain scission mechanism produces a sharp contrast between exposed and

unexposed regions, enabling the definition of features as small as 10 nanometers under optimized conditions (Chain Scission - an overview, 2018). The polymer's relatively narrow molecular weight distribution, achievable by controlled synthesis, contributes to the uniformity and reproducibility of the patterned resist film. Developer choice also plays an important role; mixtures of methyl isobutyl ketone (MIBK) and isopropyl alcohol (IPA) are commonly used to develop PMMA e-beam resists, with the MIBK:IPA ratio tuned to balance contrast and sensitivity for a given application. Although it has advantages in research settings, PMMA has several well-documented limitations that have prevented its general adoption in high-volume semiconductor manufacturing. Its sensitivity to electron beam exposure, although adequate for research applications, is relatively low compared to newer resist materials, needing increased electron doses and longer exposure times that reduce throughput. Its resistance to plasma etching used to transfer patterns into base materials is poor; PMMA etches rapidly under plasma conditions, limiting the achievable aspect ratios and pattern depths with PMMA-masked substrates. PMMA is also sensitive to environmental conditions during processing. Humidity, temperature fluctuations, and contamination can all affect film uniformity and pattern fidelity, requiring tightly controlled sterile cleanroom environments. At sub-10 nm feature sizes, line edge roughness and variations in the smoothness of patterned resist edges become major concerns, and PMMA's performance in this regard is inferior to that of state-of-the-art chemically amplified and metal oxide resist systems. These limitations have positioned PMMA primarily as a research and prototyping resist rather than a production material in modern semiconductor fabrication. Nevertheless, its role as the basic resist material against which new systems are evaluated, and its continued use in academic nanofabrication laboratories and specialized industrial applications, ensure that PMMA remains a relevant and actively studied material in the semiconductor field.

Cross-Cutting Themes and Comparative Study

The preceding sections have examined PMMA's role in biomedical and semiconductor applications independently. Taken together, however, these two application domains reveal a deeper story about the material itself, one that relates to the fundamental relationship involving polymer structure, processing, and function. Rather than representing two unrelated uses of the same chemical compound, PMMA's presence in both a surgical suite and a semiconductor cleanroom show a coherent set of material principles that translate, with appropriate modification, across radically different engineering environments. At the core of PMMA's versatility is a small set of properties that prove valuable in both application contexts, albeit for different reasons. Chemical stability: PMMA's resistance to degradation in aqueous and biological environments makes it a credible long-term implant material. In the semiconductor context, that same chemical stability is what allows a spin-coated PMMA film to survive the harsh chemical steps that precede and follow lithographic patterning without degrading prematurely. The property is identical; the environment that demands it is different. Tunable molecular weight is another cross-cutting theme. In bone cement, molecular weight governs viscosity and working time. Lower-molecular-weight formulations flow more readily, giving

surgeons adequate time to inject and position the cement before it hardens. In lithographic resists, molecular weight governs resolution and responsiveness. Higher-molecular-weight PMMA produces sharper pattern edges and finer feature definition. In both cases, engineers are manipulating the same fundamental variable to improve performance for their specific application. The underlying structure-property relationship is the same; the direction in which it is tuned differs. PMMA's susceptibility to specific organic solvents while remaining resistant to aqueous environments is also exploited in both fields. In semiconductor processing, this property underlies the development step, in which exposed PMMA dissolves cleanly in MIBK/IPA developer while unexposed regions remain intact. In biomedical research, controlled solubility is explored as a mechanism for drug delivery and for the fabrication of biodegradable scaffolds. Again, the same intrinsic property of the polymer underpins two functionally distinct applications. Despite these shared principles, the two application domains place clearly different demands on PMMA and have driven its development in very different directions. In biomedical applications, the primary engineering priorities are biocompatibility, mechanical reliability over decades of physiological loading, and biological integration. The material must coexist with living tissue without provoking harmful responses, and its mechanical function must remain stable in a warm, wet, chemically complex environment throughout the patient's lifetime.

In semiconductor applications, the primary priorities are resolution, sensitivity, and etch resistance properties that have no direct biomedical analog. The material must respond precisely to a patterning energy source, dissolve cleanly in specific regions, and survive plasma etching processes that would be irrelevant in any biological context. The timescales are also fundamentally different; a bone cement must perform for decades, while a lithographic resist film is typically stripped from the wafer within hours of deposition. These varied priorities have produced two different branches of PMMA development that share a common material foundation but have evolved largely independently. Biomedical PMMA research has focused on additives that improve biological response, hydroxyapatite fillers, antibiotic loading, and surface changes to promote osseointegration. Semiconductor PMMA research has focused on amplifying sensitivity, improving etch resistance, and engineering hybrid formulations compatible with EUV exposure. The roadmaps are different, but the starting point is the same polymer.

The Significance of Versatility

The fact that a single polymer has sustained relevance in two such different high-technology fields for several decades is not accidental. It reflects the depth and toughness of PMMA's fundamental property set, optical clarity, chemical stability, tunable molecular weight, selective solubility, and biocompatibility, which together provide enough engineering leverage to be useful across a wide range of contexts. Most materials are optimized for a narrow range of applications. PMMA's sustained importance across orthopedic surgery, dentistry, ophthalmology, nanofabrication, and photomask production speaks to an unusually broad foundation of useful properties. This versatility also provides an important lesson for materials engineers and biomedical engineers. Understanding a material at the level of its fundamental structure-property

relationships, rather than merely cataloging its applications, reveals the underlying logic that enables cross-domain utility. PMMA's versatility arises not from intentional design, but from a molecular framework that yields a broadly useful set of properties, which engineers in various fields have recognized and exploited. The insight that deep material understanding enables application discovery is a central theme in modern biomaterials science and, more broadly, in materials engineering. Despite decades of clinical and industrial use, PMMA remains an active area of research in both biomedical and semiconductor engineering. The limitations identified in Sections 4 and 5, thermal necrosis, aseptic loosening, and poor etch resistance, among others, have motivated substantial continuing efforts to extend and enhance the material's performance. At the same time, emerging technologies in both fields are creating new demands and new opportunities for PMMA-based systems. The future of PMMA is not one of replacement, but of refinement, a continued evolution of a material whose fundamental property set remains as relevant as ever. The most active area of biomedical PMMA research is the development of bioactive bone cement formulations that go beyond simple mechanical fixation to actively promote biological integration at the cement-bone interface. Incorporation of hydroxyapatite, the primary mineral component of natural bone, into PMMA cement has been shown to improve osteoconductivity, encouraging bone cells to grow toward and around the cement mantle rather than simply tolerating its presence. Similarly, the addition of bioactive glass particles has been explored to stimulate a controlled biological response that promotes bonding between the cement and surrounding tissue. These bioactive fillers represent a fundamental shift in the design philosophy of bone cement from a passive structural material to an active participant in the biological healing process. Carbon nanotube reinforcement is another emerging direction in biomedical PMMA research. Carbon nanotubes offer extraordinary mechanical properties, with tensile strengths orders of magnitude greater than steel at a fraction of the weight, and their incorporation into PMMA matrices has been shown to significantly improve fracture toughness and fatigue resistance. If the major challenges around nanotube dispersion, biocompatibility, and long-term toxicology can be resolved, carbon nanotube-reinforced PMMA cements could offer substantially improved mechanical functional longevity compared to current formulations. Antibiotic delivery represents another frontier.

Current antibiotic-loaded bone cements release their therapeutic payload over a relatively short period, with elution rates declining rapidly after implantation. Research into controlled-release mechanisms, including surface coatings, microsphere encapsulation, and polymer matrix modifications, aims to extend and better control antibiotic release profiles, thereby improving infection prophylaxis in high-risk surgical cases. Beyond antibiotics, PMMA-based systems are being investigated as means for local delivery of growth factors, anti-inflammatory agents, and other biologics to further improve the biological environment at the implant site. Reducing exothermic heat generation during polymerization remains an unresolved challenge. Approaches under investigation include the use of alternative initiator systems that polymerize at lower temperatures, pre-cooling of cement components prior to mixing, and the development of dual-setting formulations that combine free radical polymerization with slower, lower-temperature

crosslinking mechanisms. Each of these strategies aims to reduce peak interface temperatures below the threshold for thermal necrosis while maintaining adequate mechanical properties and working time.

In the semiconductor industry, the future of PMMA is closely tied to the trajectory of EUV lithography and the continuing push toward smaller feature sizes at advanced technology nodes. As transistor dimensions approach single-digit nanometers, the demands placed on resist materials become increasingly severe. Resolution, line edge roughness, and sensitivity must all improve simultaneously, a combination of requirements that no current resist material fully satisfies. PMMA hybrid resist systems in which metal oxide nanoparticles or organometallic compounds are incorporated into the PMMA matrix represent one of the most promising directions for EUV applications. These hybrid systems leverage the high EUV absorption cross-section of metal-containing compounds to improve sensitivity while retaining the resolution characteristics of the PMMA backbone. Zirconium-, hafnium-, and tin-based organometallic additives have been explored in this context, with promising results in laboratory settings. The challenge of translating these research formulations into manufacturable processes at the wafer scale remains significant. Directed self-assembly (DSA) represents another area where PMMA plays an important role. In DSA processes, block copolymers, polymer chains composed of two chemically distinct segments, spontaneously organize into regular nanoscale patterns when annealed under regulated conditions. PMMA-containing block copolymers, particularly polystyrene-block-polymethylmethacrylate (PS-*b*-PMMA), are among the most widely studied DSA materials. By selectively removing the PMMA block after self-assembly, researchers can produce nanoscale patterns with feature sizes and periodicities that complement and extend conventional lithographic techniques. DSA using PS-*b*-PMMA has been demonstrated (Chen et al., 2019) at feature sizes below 10 nm, making it one of the most promising methods for extending patterning capability beyond the limits of EUV lithography alone. Improvements in etch resistance continue as a critical research priority for semiconductor PMMA applications. Current approaches include incorporating silicon- or metal-containing additives that form a protective oxide layer during plasma etching, effectively shielding the underlying pattern from erosion. Surface treatment strategies, including plasma hardening of the resist surface prior to etching, have also demonstrated potential in extending the etch durability of PMMA-based resist films.

Converging Research Themes

Looking across both application domains, several converging themes emerge that mirror broader trends in materials engineering. In both fields, research is moving toward hybrid and composite systems that combine PMMA's established processing advantages with the functional properties of added components, whether those components are bioactive fillers, metal oxide nanoparticles, or self-assembling block copolymers. In both fields, the challenge is not replacing PMMA but augmenting it, preserving the properties that have made it necessary while handling its limitations. This convergence exhibits a broader principle in materials engineering: foundational

materials with well-understood structure-property relationships rarely become obsolete. Instead, they become platforms, starting points for composite and mixed systems that inherit their processing advantages while gaining new functional capabilities. PMMA's path in both biomedical and semiconductor engineering is a convincing illustration of this principle and suggests that the material's most important contributions may still be forthcoming.

Conclusion

Polymethyl methacrylate is, by any measure, an extraordinary engineering material. First synthesized in the early twentieth century and commercialized under trade names that have become household words, PMMA has demonstrated a lastingness and breadth of application that few synthetic polymers can match. This paper has examined the material from the ground up, tracing the relationship between its molecular framework and its macroscopic properties, exploring how those properties are modified through processing and formulation, and following the consequences of those modifications into two of the most demanding and consequential engineering fields of the modern era. The central argument of this paper has been that PMMA's presence in both biomedical and semiconductor engineering is not coincidental. It shows a coherent and robust set of structure-property relationships, including chemical stability, optical clarity, tunable molecular weight, selective solubility, and biocompatibility that provide sufficient engineering leverage to be useful across radically different application contexts. In the body, these properties enable a material that can bear physiological loads, resist biological degradation, and coexist with living tissue over the lifetime of a patient. On a silicon wafer, the same properties enable a material that can be deposited as a uniform nanometer-scale film, patterned with electron beam precision, and developed cleanly to define features at the frontier of human manufacturing capability.

The limitations of PMMA in both fields are real and well-documented. Thermal necrosis, aseptic loosening, and the absence of biological activity constrain its performance in orthopedic applications. Poor etch resistance, low sensitivity, and line edge roughness limit its role in advanced semiconductor manufacturing. But in both cases, these limitations have driven innovation rather than obsolescence, producing a rich body of research into composite formulations, bioactive additives, hybrid resist systems, and directed self-assembly techniques that extend and enhance the material's capabilities whilst preserving the processing advantages that made it valuable in the first place. For a biomedical engineer, PMMA offers a case study in translating material understanding into clinical impact. Its history, from aircraft canopy fragments to intraocular lenses, from dental prosthetics to vertebroplasty cement, illustrates how a deep understanding of structure-property relationships can unlock applications that were never anticipated at the time of the material's development. The ongoing evolution of bioactive and antibiotic-loaded PMMA formulations suggests that this process of discovery is far from complete. For a materials engineer working at the intersection of biology and technology, PMMA's dual identity is equally instructive. It demonstrates that the most valuable engineering materials are not necessarily those with the highest performance in any single metric, but those

whose fundamental property sets are broad and robust enough to be leveraged creatively across multiple domains. In this sense, PMMA is not merely a polymer; it is a platform, and one whose most significant contributions to human health and technological progress may still lie ahead.

References

- Kang, S., Lin, H., Day, D. E., & Stoffer, J. O. (2011). Optically Transparent Polymethyl Methacrylate Composites Made with Glass Fibers of Varying Refractive Index. *Journal of Materials Research*. <https://doi.org/10.1557/jmr.2010.396>
- Arash, B., Wang, Q. & Varadan, V. K. (2014). Mechanical properties of carbon nanotube/polymer composites. *Scientific Reports* 4. <https://doi.org/10.1038/srep06479>
- Geng, K. & Tsui, O. K. (2016). Effects of Polymer Tacticity and Molecular Weight on the Glass Transition Temperature of Poly(methyl methacrylate) Films on Silica. *Macromolecules*. <https://doi.org/10.1021/acs.macromol.6b00108>
- Frazer, R. Q., Byron, R. T., Osborne, P. B., & West, K. P. (2005). PMMA: an essential material in medicine and dentistry. *Journal of Long-Term Effects of Medical Implants*, 15(6), 629–639. <https://doi.org/10.1615/jlongtermeffmedimplants.v15.i6.60>
- (2014). The Effect of Nanoparticles and Alternative Monomer on the Exothermic Temperature of PMMA Bone Cement. *Materials Science and Engineering: C* 45, pp. 1-7. <https://doi.org/10.1016/j.msec.2014.09.019>
- (2022). Polymethyl Methacrylate Bone Cement Polymerization Induced Thermal Necrosis at the Cement–Bone Interface: A Narrative Review. *MDPI* 14(24). <https://doi.org/10.3390/app142411651>
- Chain Scission – an overview. (2018). In *Handbook of Material Weathering* (6th ed.). Elsevier. <https://www.sciencedirect.com/topics/engineering/chain-scission>
- Chen, S., Wu, G., Wang, X., Chen, X., & Nealey, P. (2019). Sub-10 nm feature sizes of disordered polystyrene-block-poly(methyl methacrylate) copolymer films achieved by ionic liquid additives with selectively distributed charge interactions. *ACS Applied Polymer Materials*, 2(2). <https://doi.org/10.1021/acsapm.9b00898>
- (2022). Optical and thermal properties of polymethyl methacrylate (PMMA) bearing phenyl and adamantyl substituents. <https://doi.org/10.1016/j.jlumin.2022.118013>
- Chen, S., Wu, G., Wang, X., Chen, X., & Nealey, P. (2019). Sub-10 nm feature sizes of disordered polystyrene-block-poly(methyl methacrylate) copolymer films achieved by ionic liquid additives with selectively distributed charge interactions. *ACS Applied Polymer Materials*, 2(2).
- Placek, L., Perry, D., Towler, M., & Wren, A. (2025). Gallium-containing bioactive glasses: Their influence on ion release and the bioactivity of resulting glass polyalkenoate cements. *Applied Sciences*, 15(14), 7756.

